

## Chapter 2

# Basic Notions

## 2.1 Sets

A **set** is a collection of elements. We denote sets with curly brackets  $\{\dots\}$ , with the elements listed within the brackets. As an example, consider the set of students in the class *Mathematical Analysis 1*. This set can either be written explicitly,

$$X = \{\text{Bob}, \text{Lucy}, \text{Andrew}, \text{Giulia}\}$$

or it can be defined using a rule:

$$X = \{\text{all people who are students of } \textit{Mathematical Analysis 1}\}.$$

Some important sets of numbers that we will often encounter are

$$\mathbb{N} = \text{set of natural numbers} = \{0, 1, 2, \dots\}$$

$$\mathbb{Z} = \text{set of integer numbers} = \{0, \pm 1, \pm 2, \dots\}$$

$$\mathbb{Q} = \text{set of rational numbers} = \{\text{can you think how to define?}\}$$

$$\mathbb{R} = \text{set of real numbers} = \{\text{can you think how to define?}\}$$

$$\mathbb{C} = \text{set of complex numbers} = \{\text{can you think how to define?}\}$$

Another important set is the **empty set** which contains no elements. It is denoted  $\emptyset$ .

### Basic notation

- **Element of:** if  $x$  is an element of  $X$  we write  $x \in X$
- **Not element of:** if  $x$  is *not* an element of  $X$  we write  $x \notin X$
- **Subset:** if  $A$  is a subset of  $X$  (i.e. any element of  $A$  is also an element of  $X$ ) we write  $A \subseteq X$  or  $X \supseteq A$   
In this case it is possible that  $A = X$ .
- **Proper subset:** if  $A$  is a *proper* subset of  $X$  we write  $A \subsetneq X$  or  $X \supsetneq A$   
In this case  $A \neq X$  (i.e. there exists  $x \in X$  such that  $x \notin A$ ).

**Lemma 2.1:** For some set  $X$  and subsets  $A, B \subseteq X$ , if  $A \subseteq B$  and  $B \subseteq A$  then  $A = B$ .

*Proof.* By contradiction, assume that  $A \neq B$ . Then, without loss of generality, there exists  $x \in X$  such that  $x \in A$  and  $x \notin B$ . But then it is not true that  $A \subseteq B$ . The contradiction assumption must therefore be false, i.e.  $A = B$ .  $\square$

### Characteristic property

The elements of a subset  $A \subseteq X$  can often be characterized by a mathematical property that they satisfy. This property is denoted  $p(x)$ , and we write

$$A = \{x \in X \mid p(x)\}.$$

For example, if  $p(x) = 'x \text{ is even}'$ , then

$$A = \{x \in \mathbb{N} \mid x \text{ is even}\} = \{0, 2, 4, \dots\} \subseteq \mathbb{N}.$$

### Operations on sets

- **Complement:** if  $A \subseteq X$  then we define its *complement* to be

$$A^C = \{x \in X \mid x \notin A\}.$$

- **Union:** for two sets  $A \subseteq X$  and  $B \subseteq X$  we define their *union* to be

$$A \cup B = \{x \in X \mid x \in A \text{ or } x \in B\}$$

- **Intersection:** for two sets  $A \subseteq X$  and  $B \subseteq X$  we define their *intersection* to be

$$A \cap B = \{x \in X \mid x \in A \text{ and } x \in B\}$$

- **Difference:** for two sets  $A \subseteq X$  and  $B \subseteq X$  we define their *difference* to be

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\}$$

- **Disjoint Union:** for two sets  $A \subseteq X$  and  $B \subseteq X$  whose intersection is empty, we often replace the symbol  $\cup$  by

$$A \sqcup B$$

**Lemma 2.2** (Properties of  $\cap$  and  $\cup$ ): For some set  $X$  and subsets  $A, B, C \subseteq X$  the operations  $\cap$  and  $\cup$  satisfy:

1. Boolean properties:  $A \cap A^C = \emptyset$  and  $A \cup A^C = X$ .
2. Commutativity:  $A \cap B = B \cap A$  and  $A \cup B = B \cup A$ .
3. Associativity:  $(A \cap B) \cap C = A \cap (B \cap C)$  and  $(A \cup B) \cup C = A \cup (B \cup C)$ .
4. Distributivity:  $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$  and  $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ .